

Inputs

Grain and fluid moduli
SAG-P2/P3 log medians
CarbFix and Wallula
field constraints

Rock physics

Cementation law $f(t)$
Porosity loss and
frame stiffening
Reuss/patchy + Gassmann

Uncertainty

Stiffness, end member
 f_{∞} - τ sweep
Monte Carlo ($N = 20\,000$)
 V_p vs V_s ambiguity

2-D monitoring

Reservoir 4×1.6 km
Convolutional 4-D
Noise test, time shifts
Synthetic well logs